

UQFF 2027 Decisive-Experiment Triple: Joint P6+P11+P12 Falsification Geometry

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1. Setup

Three independent decisive experiments converge in 2027:

- **P6** (PAPER_1173): Wuhan torsion-balance sub-mm Yukawa, expected $L_{\text{KK}}^* = 26.3 \mu\text{m}$.
- **P11** (PAPER_1175): LIGO O5 stacked ringdown, expected $\mathcal{R}_{\{21/22\}} = 0.144 \pm 0.010$.
- **P12** (PAPER_1176): Euclid Y1 weak lensing, expected $\sigma_8 = 0.797 \pm 0.005$.

All three originate from the same R26 closed Lagrangian (PAPER_1167--1173) and share a single geometric scale: $D_{\text{crit}}/D_{\text{BSFG}} = 13/3 \approx 4.333$.

2. Joint Closed Form

Each prediction observable can be written as a function of the single geometric ratio

$\xi = D_{\text{crit}}/D_{\text{BSFG}}$:

$$\begin{aligned} L_{\text{KK}}^*(\xi) &= \hbar c / m_1(\xi), \quad m_1^* \propto \xi^{-1} \\ \mathcal{R}_{\{21/22\}}(\xi) &= 0.10 \cdot \xi^{1/4} \\ \sigma_8(\xi) &= 0.7851 \cdot (1 + 0.01525 \xi/\xi_0), \quad \xi_0 = 13/3 \end{aligned}$$

All three lock at the canonical value $\xi_0 = 13/3$ to give the central UQFF predictions.

3. Joint Likelihood Geometry

For a hypothetical 2027 measurement triple $(\hat{L}, \hat{\mathcal{R}}, \hat{\sigma}_8)$ with 1σ widths $(\sigma_L, \sigma_R, \sigma_8^\sigma) = (5 \mu\text{m}, 0.012, 0.005)$, the joint χ^2 in the single-parameter ξ -space reduces to

$$\begin{aligned} \chi^2(\xi) &= \left(\frac{L_{\text{KK}}^*(\xi)}{\sigma_L} - \frac{\hat{L}}{\sigma_L} \right)^2 \\ &+ \left(\frac{\mathcal{R}_{\{21/22\}}(\xi)}{\sigma_R} - \frac{\hat{\mathcal{R}}}{\sigma_R} \right)^2 \\ &+ \left(\frac{\sigma_8(\xi)}{\sigma_8^\sigma} - \frac{\hat{\sigma}_8}{\sigma_8^\sigma} \right)^2. \end{aligned}$$

UQFF survives if $\min_{\xi} \chi^2(\xi) < 7.81$ (3-dof, 95% CL).

4. Cross-Constraint Predictions

If any single experiment returns a value off the UQFF prediction, the other two constrain ξ :

Hypothetical 2027 outcome	Implied ξ	UQFF status
$L_{KK}^2 = 26 \mu m$, $R = 0.144$, $\sigma_8 = 0.797$	13/3	*CONFIRMED*
$L_{KK}^2 = 50 \mu m$	8.27	EXCLUDED (Euclid forces $\xi < 4.5$)
$\sigma_8 = 0.815$	$\sigma_8 > 5.5$	EXCLUDED (Wuhan + LIGO force $\xi < 5.0$)
$R = 0.105$	1.21	EXCLUDED (Wuhan forces $\xi > 4.0$)

The triple is mutually self-locking: **no single measurement can rescue UQFF if either of the other two falsifies it.**

5. Joint Falsification Decision Tree

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..
2027 Wuhan
Yukawa @ 20 um?
/\
/ \
YES / \ NO
/ \
UQFF P6 UQFF P6
CONFIRMED FALSIFIED
/ |
2027 LIGO 05 v
R_21/22 in range? UQFF R26 EXCLUDED
/\ (regardless of P11, P12)
/ \
YES / \ NO
/ \
Euclid UQFF P11
Y1? FALSIFIED
/
sigma_8 in range?
/\
YES NO
| \--> UQFF P12 FALSIFIED
v
FULL UQFF R26 CONFIRMED
(all 3 of 3 within 1-sigma)
..

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6. Cross-Check with CP4 #261

CP4 calculator UQFF2027JointFalsifierCalculator accepts hypothetical $(\hat{L}, \hat{R}, \hat{\sigma}_8)$ and returns posterior ξ , $\chi^2_{\min}$, status.

7. Status

CLOSED. PAPER_1177 unifies P6/P11/P12 into a single ξ -parameter falsification geometry.